

GAMING CATEGORY MID-RANGE VIDEO CARDS

The 'Mid' category of cards was the smallest in terms of numbers: just three. The mid-level cards this year consisted of the high-end cards from last year. There were two ATi, both 9800 PRO's, from Club3D and PowerColor respectively, and one Gainward GeForce FX 5900 XT. With not much left to compare among the ATi's, it was finally performance that mattered.

Their Features

On the features front, the cards were plain vanilla with not much to talk about. With the regular bundle of old games and driver software along with cables and DVI adapter, formed the package of all three cards.

Their Performance

The clock speed is one thing that needs to be talked about here. Both 9800s were clocked at the same speed—which is a default 380MHz—the 5900XT, though, was clocked higher in terms of clock speed and memory speed. The pixel pipelines, however, are where the tables turned.

The 5900XT is more in the league of the 9600XT than the 9800PRO. It has four pixel pipelines as a standard but it can do pseudo eight-pixel pipeline operations. This, in non-technical language, means that it renders four conventional (colour + Z-order) pixels per clock, but is

capable of performing eight operations per clock for Z pixels, textures, and stencil and shader ops, which makes it partially crippled. Also, it is a toned-down version of the 5900 Ultra in terms of clock speed and is placed just above the 5700 Ultra in terms of price and the difference is around Rs 250. This card is also termed as the 5900 SE in some cases.

In our Tier 1 tests, the Gainward 5900 XT blazed through the games. It gave more than acceptable frame-rates at high resolutions across all categories of games. The Club3D 9800 PRO 128MB made it through to the Tier-2 tests along with the Gainward GeForce FX 5900XT.



In the Tier-2 tests, where the in-game settings were turned on to high, the Gainward reigned supreme. But it could not come even close to the 9800 in the *Call of Duty* tests, which left it biting the dust. In all other

The 5900XT gives you the most bang for your buck

games, though, the ATi 9800 was second-best. Once again, we encountered the same problem in *Far Cry* at 1280 x 1024 where the card posted a measly frame-rate of 5 fps with visible artefacts. This did the card in all the more and the 5900XT came through to win this competition.

The Winner Is...

The 5900XT is a strange species. With its pseudo eight-pixel pipeline processing, it could have fit into the 'Low' category, and would have beaten the daylights out of all the other cards including the 5700 Ultra.

On the other hand, the pseudo processing feature also made it a candidate for the mid-level category and that's where it went in the end. But the performance that it offered was more than satisfying and it unquestionably gives you more bang for your buck. The Gainward 5900XT 128 MB is undoubtedly the winner of the Digit Best Buy Gold Award in this category.

GAMING CATEGORY HIGH END VIDEO CARDS

In the comparison we carried last year, the 9800 PRO and the 5900 Ultra were the two top-of-the-line cards from ATi and nVidia respectively. Both cards had the 8-pixel pipeline architecture, running 256 MB of DDR RAM.

Cut back to the present, and we have the X800XT PE and the 6800 Ultra from the same manu-

OEM Deals And How They Affect Your Gaming Experience: Doom 3 & Half-Life 2

Two of the most awaited games this year were *Doom 3* and *Half-Life 2*, which we used for our benchmarks. Most gamers around the planet swear by the lighting effects used in *Doom 3* and the water detail in *HL2*. However, are you sure that the card you have will give you the best visual experience when you play these games?

Doom 3 by far is one of the most taxing games when played with the High settings. Both nVidia and ATi cards give comfortable frame-rates when played at lower resolutions. As you crank up the resolution, the nVidia cards just zoom past the top-end ATi's, giving a performance increase of more than 30 per cent. So does that mean that ATi cards cannot play *Doom 3* well? There is some truth to this. The ATi drivers are comparatively bad on OpenGL implementation vis-à-vis nVidia.

Another reason is that nVidia's 6-series architecture is optimised precisely for games such as *Doom 3*, since they worked more closely with the ID team during the development of the game. It is clear from this that any game based on the *Doom 3* engine would most probably favour nVidia cards in the future.

On the other hand, there is *Half-Life 2*. Similar to all games, when you start the game it auto-detects the video card and provides you with a setting that gives you the best in visual quality for the card that you have. Fair enough, but if you check the Advanced Settings in the Video menu option, you may be in for a surprise. With the nVidia 5-series video cards, the DirectX option is set to 8.1 by default in the game. But with the 9-series ATi, the default DirectX level chosen is 9. Surprised? We were too.

After some research we got the real deal. Apparently, Valve designed the game to use 24-bit floating point instructions at all times in the game. ATi has 16-bit and 24-bit floating-point registers while the 5-series NVIDIA has 16 and 32 bit registers, but no 24-bit ones. Since 16-bit is partial precision it can be used both in DX9 as DX 8.1 implementations. As the 24-bit registers are absent in the nVidia card, the game checks for them, doesn't find them, and automatically implements DirectX 8.1 instead of DirectX 9! However, there is a way to run the 5-series cards in DirectX 9 in *HL2*. For this, start the console in *HL2*, and there, type in "dx_matlevel 90" without the quotes. This will make the card run the game in DirectX 9 at all times.

Oh, and before we forget, ATi has an OEM deal with Valve for *Half-Life 2*!